Fuzzy Data Analysis

Fuzzy Analysis of Delivery Outcome

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1. Introduction

Cardiotocography (CTG) is one of the basic methods of biophysical monitoring of the intrauterine fetal development. It involves the simultaneous recording and analysis of fetal heart rate (FHR) signal, fetal movement activity and uterine contractile function. The correct interpretation of the FHR signal is of special importance as it is the most crucial for the reliable assessment of the fetal state. Computational intelligence methods and in particular procedures based on the supervised learning approach may be distinguished among the most efficient methods of automated qualitative FHR signal analysis. Classic examples are artificial neural networks, support vector machines, fuzzy classifiers, random forests, etc.

During the supervised learning the optimal classifier parameters are estimated using an appropriate set of exemplary data (the training set). The machine based fetal state assessment requires a training data consisting of the parameters quantitatively describing the CTG signals and the corresponding reference interpretation. There are two basic approaches leading to determination of the reference fetal assessment. In the first, the reference signals interpretation is provided by a clinical expert. In the second, the delivery outcome is retrospectively assigned as the fetal state during the time of the monitoring. However, as the qualitative evaluation of the CTG signals by clinicians is highly subjective [7], the method for determining the retrospective fetal state assessment based on the fuzzy analysis of the delivery outcome attributes can be applied [3]. The result of fuzzy inference allows also for choosing the signals related to the fetal distress that are characterized with the highest diagnostic value – the highest fuzzy score.

2. Retrospective fetal state assessment

The qualitative assessment of the CTG signals involves assigning it to one of two classes, defining the fetal state as normal or abnormal. Several studies distinguish also the third, indecisive class called suspicious or questionable, that includes the cases which require further investigation. There is no other noninvasive diagnostic method, which allows for confirming the real fetal state at the time of the fetal monitoring session. In perinatology it is assumed however, that the fetal state can not change rapidly during pregnancy [6]. Hence, the results of the delivery outcome assessment can be used as the reference to verify the efficiency of the automated classification. Although this approach is not entirely correct as the abnormal outcome can be also the result of complications during a labour, it still remains the most objective method of determining the reference fetal state interpretation for automated classification algorithms.

The delivery outcome can be determined based on the analysis of the three main attributes evaluated by clinicians: the Apgar score (AP) representing the result of the visual assessment of the newborn, the birth weight (BW) expressed in percentiles in relation to a given population of newborns and the pH measurement of the acid-base balance of the umbilical cord blood (PH). For each of these attributes there are ranges of values related to the particular classes of the delivery outcome (Table 1). The supervised learning is usually carried out on the basis of a single attribute values

[5, 2]. Consequently, the information of the delivery outcome, which can be obtained as a result of the simultaneous analysis of all attributes is lost. Hence, the fuzzy system for determining the retrospective assessment of the fetal state based on the analysis of all the three attributes can be applied. The results of the fuzzy inference (the fuzzy score) provides also the diagnostic value of the final assessment of the reference fetal state which can be used to improve the efficiency of the supervised learning.

Table 1: The class labels of the delivery outcome attributes

Delivery outcome	AP	Attribute BW	PH
Normal	≥ 7	≥ 10	≥ 7.2
Suspicious	$[5, 6]$	$(5, 10)$	$[7.1, 7.2]$
Abnormal	< 5	≤ 5	< 7.1

3. Fuzzy analysis

The fuzzy inference is conducted on the basis of fuzzy conditional rules with three inputs, being the values of outcome attributes, and one output representing the final delivery outcome:

$$\underset{1 \leq i \leq I}{\forall} R^{(i)} : \bigwedge_{j=1}^{3} \left(x_{0j} \textbf{ is } A_j^{(i)} \right) \textbf{ then } y^{(i)} = y_0^{(i)}, \tag{1}$$

where: I is the number of rules, x_i denotes inputs (Apgar score (x_{01}), birth weight (x_{02}), pH (x_{03})), and $y^{(i)}$ is the output. Symbol $A_j^{(i)}$ denotes the linguistic value representing the class of the delivery outcome being the result of the assessment of a single outcome attribute. It can be defined by a fuzzy set with trapezoid membership function $\mu_{A_j}^{(i)}(x)$. However, Gaussian and sigmoidal functions can also be used. The parameters of trapezoid membership functions can be determined using simple statistical analysis of delivery attributes dataset [1]. The core of the fuzzy set (c, b) (see Figure 1) can be defined as the interquartile range of parameters values of the particular class of delivery outcome. The limit values (a, d) can be calculated to fulfill the assumption that the membership of the values defining the boundary between classes should be the same and equal to $\mu_{A_j}^{(i)}(x) = 0.5$ for both classes. Similar approach can be applied in the case of Gaussian and sigmoidal functions.

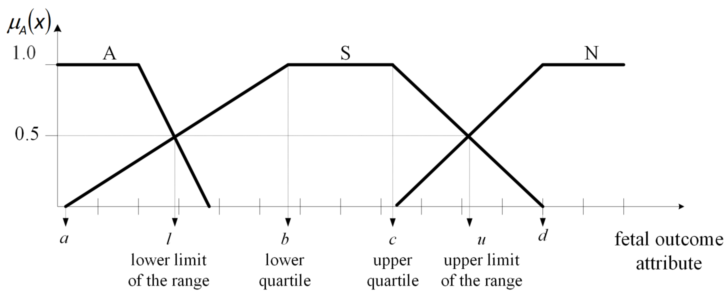

Figure 1: An example of membership functions defining possible ranges of a fetal outcome attribute.

The rule base (1) defines a zero-order Takagi-Sugeno-Kang (TSK) fuzzy model [8, 4]. The consequence $y^{(i)}$ is a fuzzy set with the singleton membership function. Its location $y_0^{(i)}$ is defined by assuming that a normal outcome corresponds to negative $y_0^{(i)} = -1$, whereas the abnormal to positive output value $y_0^{(i)} = +1$ of a single fuzzy rule. To classify the delivery outcome as suspicious a varying location is assumed $y_0^{(i)} = p^{(i)}$ (Figure 2), allowing for a different interpretation of delivery outcome attributes from the ranges corresponding to the suspicious class.

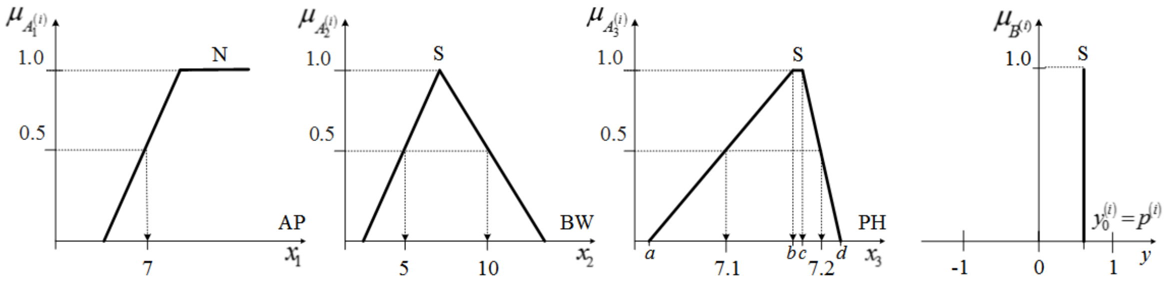

Figure 2: An example of a fuzzy rule indicating the suspicious delivery outcome. As only one attribute is within the normal range and there are no attributes related the abnormal state, the fuzzy rule output defines the suspicious delivery outcome.

The proposed TSK model is equivalent to Mamdani fuzzy reasoning system with the consequences of fuzzy sets in the form of isosceles triangular membership functions with the width of the base equal to 2 and the center of gravity defuzzification.

A complete rule base of the TSK fuzzy system consists of $I = 3^3 = 27$ conditional statements. In the proposed solution, the delivery outcome is interpreted as:

- abnormal, if any of attributes indicates the abnormal state,
- normal, if two or more attributes indicate the normal and none indicates the abnormal state,
- suspicious, for all the remaining cases.

An example of a fuzzy rule indicating the suspicious delivery outcome is shown in Figure 2.

The final output value of the fuzzy model is calculated as a weighted mean of all rules outputs:

$$y_0 = \frac{\sum_{i=1}^{I} \mu_{\mathbf{A}^{(i)}}\left(\mathbf{x}_0\right) y_0^{(i)}}{\sum_{i=1}^{I} \mu_{\mathbf{A}^{(i)}}\left(\mathbf{x}_0\right)}, \tag{2}$$

where: $\mu_{\mathbf{A}^{(i)}}\left(\mathbf{x}_0\right) = \prod_{j=1}^{3} \mu_{A_j}^{(i)}\left(x_{0j}\right)$.

The above equation can be interpreted as an ensemble of experts, each represented by a single rule. The fuzzy rule defines the relationship between particular outcome attributes and the class of the final delivery outcome assessment. The weighted average of the assessments from all rules (experts) formulates the final interpretation of the delivery outcome.

In the further considerations, we assume two classes of the delivery outcome assessment - normal and abnormal. Consequently, a delivery outcome is classified as abnormal if the positive output value (2) exceeds a predefined threshold $y_0 \geq \Delta$. When considering a three class assessment problem the suspicious state can be defined as corresponding to $|y_0| < \Lambda$, where Λ is the limit threshold value.

4. Tasks

Task 1. For the dataset indicated by a teacher, determine the membership functions (the type of function will be specified for each section), which define the rule base of the TSK system to determine the delivery outcome.

Task 2. Using the grid search method, find the inference system parameters: $p^{(i)}$ set to be the same for all rules defining a suspicious fetal condition and Δ, which provides the highest correspondence between the binary fuzzy score and the assessment of the fetal outcome defined by pH measurements only (pH limits indicating the abnormal delivery outcome will be provided by a teacher). The parameters $p^{(i)}$ and Δ of the fuzzy model should be varied in the range $[-0.50, +0.50]$ with a step of 0.25. What is the value of the mean G-measure of the fetal outcome classification calculated from the 5-fold cross-validation procedure.

Task 3. Determine $p^{(i)}$ (separately for each rule defining a suspicious fetal condition) and the Δ providing the highest value of the G-measure of the classification using evolutionary strategy. Select the best performing parameters by a 5-fold cross-validation. How many signals are characterized by the high informativeness level $|y_0| > 0.5$ using system parameters from the best fold?

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